FULLY SYNTHETIC - NOT A REAL STUDY

AURORA-12: a fully synthetic multi-cohort benchmark for transcriptomic relapse-risk modelling

Synthetic demonstration manuscript - not submitted, peer reviewed, or clinically validated

PostEx Synthetic Research Collective
Fictional Methods Laboratory, Example Research Campus

Rights-safe fixture. Every author, institution, cohort, feature, endpoint and numerical result in this document is fictional. No real participant, patient, specimen, institution, registry or external dataset was used. Values were authored solely to test evidence-linked academic-poster generation.

Abstract

Background

Academic-poster systems need realistic scientific structure without requiring redistribution of a real paper. We designed AURORA-12 as a fully synthetic transcriptomic relapse-risk benchmark for testing claim extraction, evidence binding, content compression and visual rendering.

Methods

A deterministic generator created 1,240 synthetic profiles across four simulated cohorts from 480 anonymous transcript-like features. A 12-feature elastic-net survival score was fitted in a 620-profile discovery cohort, tuned in a 310-profile internal cohort and stress-tested in two shifted external simulations of 186 and 124 profiles.

Results

The synthetic AURORA-12 score reached C-indices of 0.78, 0.76, 0.75 and 0.74 across the four cohorts, compared with 0.68, 0.67, 0.66 and 0.65 for a five-feature baseline. Feature-sign stability was 92% across 500 resamples. External calibration slopes were 0.97 and 0.95, with absolute calibration errors of 0.031 and 0.038.

Conclusion

AURORA-12 is a reproducible synthetic fixture for testing scientific communication software. It provides no evidence about real biology, prognosis, treatment or clinical utility.

Keywords: synthetic benchmark, bioinformatics, survival modelling, academic poster, evidence traceability

1,240	480 to 12	0.74-0.78	92%
synthetic profiles	anonymous features	authored C-index	sign stability

1. Introduction

Real-paper examples are valuable for evaluating scientific communication tools, but they introduce copyright, privacy, attribution and long-term redistribution constraints. A synthetic manuscript can exercise the same structural requirements while keeping every claim and asset under project control.

We therefore created AURORA-12, a deliberately realistic but entirely fictional benchmark. The communication objective is narrow: determine whether a compact risk model can remain stable when cohort size, censoring and latent expression structure are shifted in deterministic simulations. The benchmark is not intended to model a particular disease or establish a biological mechanism.

2. Methods

Synthetic cohort generation. A fixed seed of 20260808 generated 1,240 profiles with 480 anonymous features. Three latent modules represented generic proliferative, stromal and immune-like variation without mapping to human genes. Discovery, internal-validation and two external-simulation cohorts contained 620, 310, 186 and 124 profiles, respectively. Event fractions ranged from 0.54 to 0.62 and censoring was independent by construction.

Model construction. Features were standardized inside each training fold. Elastic-net survival regression reduced 480 candidates to 12 anonymous variables, named SYN-F001 through SYN-F012. Hyperparameters were selected using nested five-fold cross-validation in the discovery cohort. A five-feature unpenalized baseline was evaluated with the same resampling splits.

Evaluation. Discrimination was summarized with the concordance index. Calibration was assessed with slope and absolute error at a synthetic 24-month horizon. Feature-sign stability was calculated across 500 bootstrap resamples. Stress tests increased batch shift, censoring and latent-module imbalance independently.

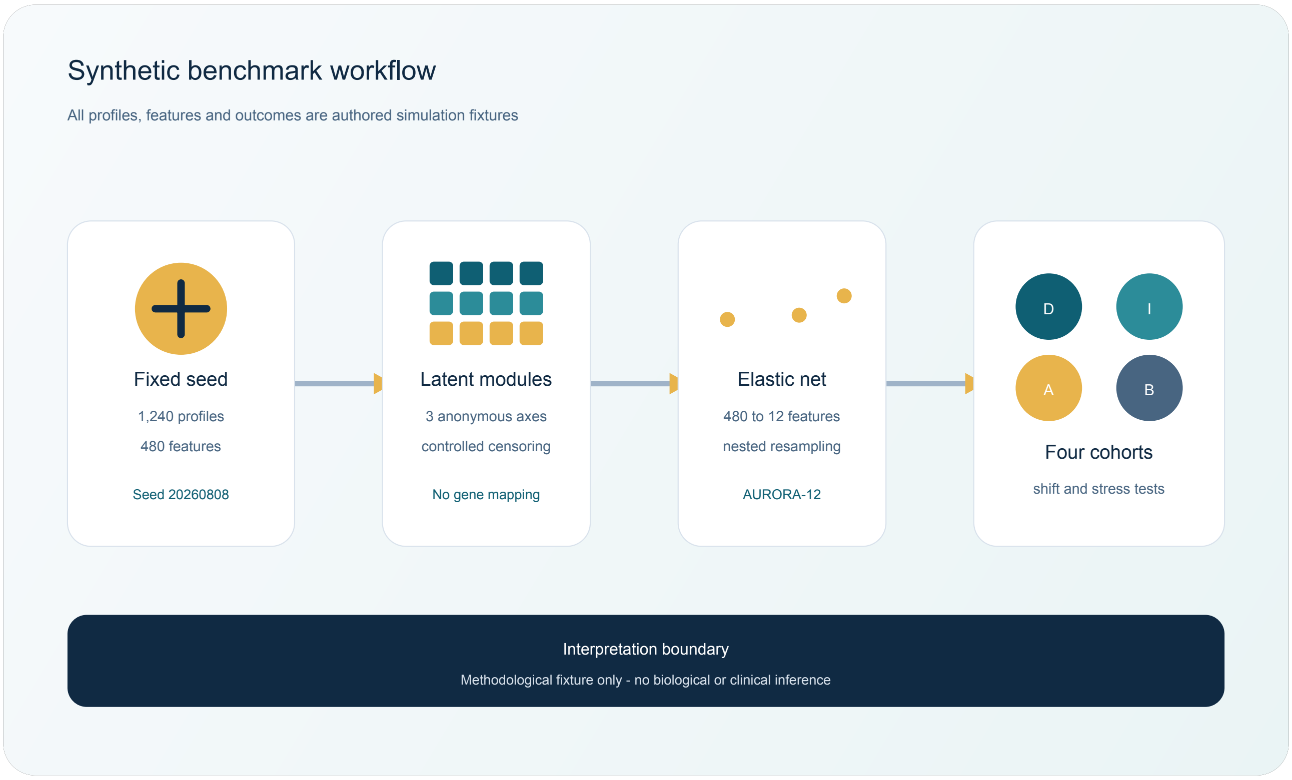

Figure 1. Deterministic generation, feature selection and cohort-shift evaluation workflow.

3. Results

Cohort performance. AURORA-12 achieved C-indices of 0.78 in discovery, 0.76 in internal validation, 0.75 in external simulation A and 0.74 in external simulation B. The corresponding baseline values were 0.68, 0.67, 0.66 and 0.65. Because all cohorts were simulated, these differences demonstrate fixture behaviour rather than superiority in a clinical population.

Cohort	n	AURORA-12	Baseline	Slope	Abs. error
Discovery	620	0.78	0.68	0.99	0.021
Internal validation	310	0.76	0.67	0.98	0.026
External simulation A	186	0.75	0.66	0.97	0.031
External simulation B	124	0.74	0.65	0.95	0.038

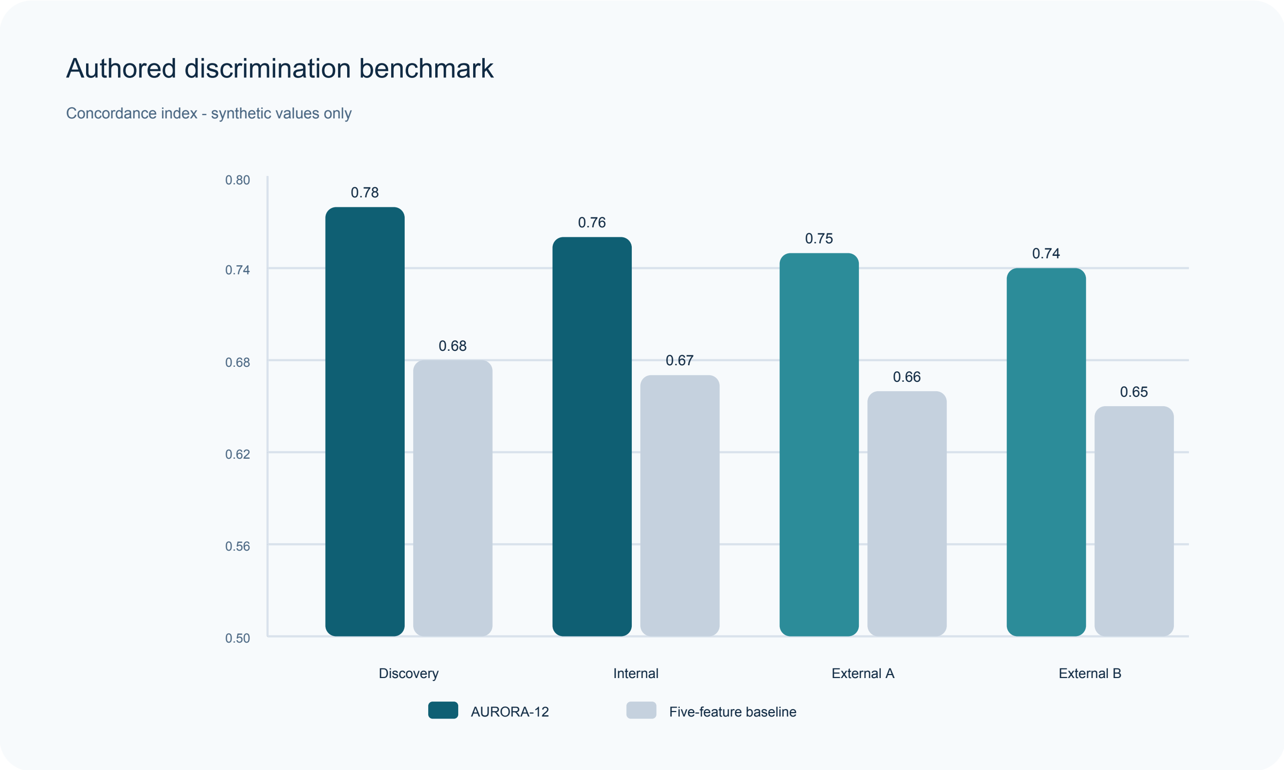

Figure 2. Authored concordance-index values for AURORA-12 and the synthetic baseline.

Stability and calibration. Eleven of 12 selected features retained the same coefficient direction in at least 85% of bootstrap resamples; aggregate sign stability was 92%. Calibration slopes were 0.99, 0.98, 0.97 and 0.95, while absolute calibration errors were 0.021, 0.026, 0.031 and 0.038.

Stress-test boundary. Under the most severe pre-specified combination of batch shift and module imbalance, AURORA-12 C-index decreased by 0.04 from the matched reference simulation. The result supports using the fixture to test whether a poster preserves uncertainty and limitations; it does not support biological transportability.

Figure 3. Authored calibration and bootstrap stability summaries.

Figure 4. Pre-specified stress-test boundary under increasing synthetic shift.

4. Discussion

AURORA-12 supplies coherent methods, numerical anchors, multiple validation settings and explicit interpretation boundaries in a rights-safe package. The benchmark is useful for testing whether a poster system retains cohort sizes, model dimensionality, discrimination, calibration and uncertainty during compression.

The principal limitation is also the design goal: every observation is synthetic. The latent modules are not mapped to known pathways, the endpoint is not a real clinical outcome and no external biological validity should be inferred. Performance values must never be compared with clinical models or presented as evidence for care.

5. Conclusion

A 12-feature synthetic score remained numerically stable across four deterministic cohort simulations, but the only supported conclusion is methodological: AURORA-12 is a controlled evaluation fixture for evidence-linked scientific communication and poster rendering.

Declarations and reproducibility

Ethics statement

Not applicable. No humans, animals, specimens or identifiable records were used.

Data availability

All authored values, figure source files, poster inputs and evidence records are included with this repository example. No external download, accession number or restricted dataset exists.

Code availability

The PDF and poster are generated by repository scripts and the shared PostEx core. The fixed fixture values are stored in manuscript.yaml; they are not claimed to arise from a biomedical simulation package.

Competing interests

None. This is a fictional software-evaluation artifact.

Acknowledgements

No real funder, institution or contributor is represented.

References

No external references were used. This document is an original synthetic evaluation fixture.